

# MANUEL RODRÍGUEZ VILLEGAS

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[manuel-rodriguezvillegas.github.io](https://manuel-rodriguezvillegas.github.io)

## PROFILE

AI engineer with international research and industry experience and a strong mathematical background. My approach starts with first-principles reasoning to turn complex problems into simple, efficient solutions. I bring energy and enjoy fast-paced environments where I can take ownership and meaningful responsibility.

## EDUCATION

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| <b>Comillas Pontifical University, ETSI ICAI</b><br><i>Master's Degree in Artificial Intelligence; GPA: 9.08/10; Class Representative</i><br>– Ongoing Master's thesis: Vision-only autonomous drones with differentiable physics. | Madrid, Spain<br>2025 – 2026      |
| <b>Cornell University</b><br><i>Exchange Student - Electrical &amp; Computer Engineering</i>                                                                                                                                       | Ithaca, NY, USA<br>Jan – May 2025 |
| <b>Comillas Pontifical University, ETSI ICAI</b><br><i>BE in Mathematical Engineering and AI; GPA: 8.75/10</i>                                                                                                                     | Madrid, Spain<br>2021 – 2025      |

## EXPERIENCE

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| <b>AUDI AG</b><br><i>Geometric AI Intern</i><br>– Developing 3D defect classification under limited data and strong confounding.<br>– Building AI agent skills for NLP-based requirements classification and conflict detection, and reliable, low-latency, and token-efficient access to a graph database with 100,000+ records. | Ingolstadt, Germany<br>July 2026 – Nov 2026 |
| <b>Comillas Pontifical University</b><br><i>Teaching Assistant</i><br>– Contributed to deployment of a GPU-based AI cluster to support student workloads in deep learning courses.                                                                                                                                                | Madrid, Spain<br>Sep 2025 – Apr 2026        |
| <b>Azzulei Technologies</b><br><i>Computer Vision Intern</i><br>– Built a real-time multi-object tracking pipeline (YOLOv8 + Optical Flow) for automated sports camera control, achieving robust performance at 30 FPS.                                                                                                           | Madrid, Spain<br>Jun – Aug 2025             |
| <b>Imperial College London</b><br><i>Research Assistant</i><br>– Reproduced nonlinear dynamical systems using Neural ODEs and PINNs (PyTorch), validating stability and trajectory accuracy against ground-truth simulations and different initial conditions.                                                                    | Remote (London, UK)<br>Jun – Aug 2024       |
| <b>Endesa</b><br><i>Energy Data Analyst Intern</i><br>– Forecasted daily Iberian ancillary service costs using MLP and regression models (MAPE < 10%).                                                                                                                                                                            | Madrid, Spain<br>Jun – Aug 2023             |

## SELECTED RESEARCH

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| <b>TopoSIGMA: Spectral Information-Geometric Mapping Architecture</b><br>– Designed an online, training-free ROS 2 mapping system separating spectral place discovery from covariance-aware loop closure, using DINOv2 descriptors and directional node distributions.<br>– Manuscript in preparation for submission. | 2026 |
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## SKILLS

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| <b>Software:</b> Python (advanced), PyTorch, JAX, ROS 2, Git, Docker, K8S, LangGraph, n8n                        |
| <b>AI &amp; ML:</b> Deep Learning, Generative & Agentic AI (MCP), Geometric AI, Computer Vision, NLP, MLOps, DRL |
| <b>Mathematics:</b> Linear Algebra, Probabilistic AI, Optimization, Differential Geometry, Statistics            |

## LANGUAGES

Spanish (Native) | English (Fluent) | German (Beginner)

## HONORS & AWARDS

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|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Bending Spoons Hackathon Winner</b><br><i>Selected among Spain's top 20 tech students for First Ascent and winner of the event's hackathon.</i> | Milan, Italy   2026              |
| <b>Hackathon Winner</b><br><i>Developed the winning project in the 10th Smart Industry Hackathon, featuring Kearney and CAF.</i>                   | Madrid, Spain   2025             |
| <b>UNIJES Social Tech Challenge Winner</b><br><i>Voice-controlled wheelchair project</i>                                                           | Bilbao, Spain   2024             |
| <b>Academic Excellence Scholarship</b><br><i>Top-scoring students across Community of Madrid</i>                                                   | Madrid, Spain   2021, 2023, 2024 |
| <b>Baccalaureate Honorable Mention</b><br><i>2nd highest GPA in Community of Madrid</i>                                                            | Madrid, Spain   2021             |